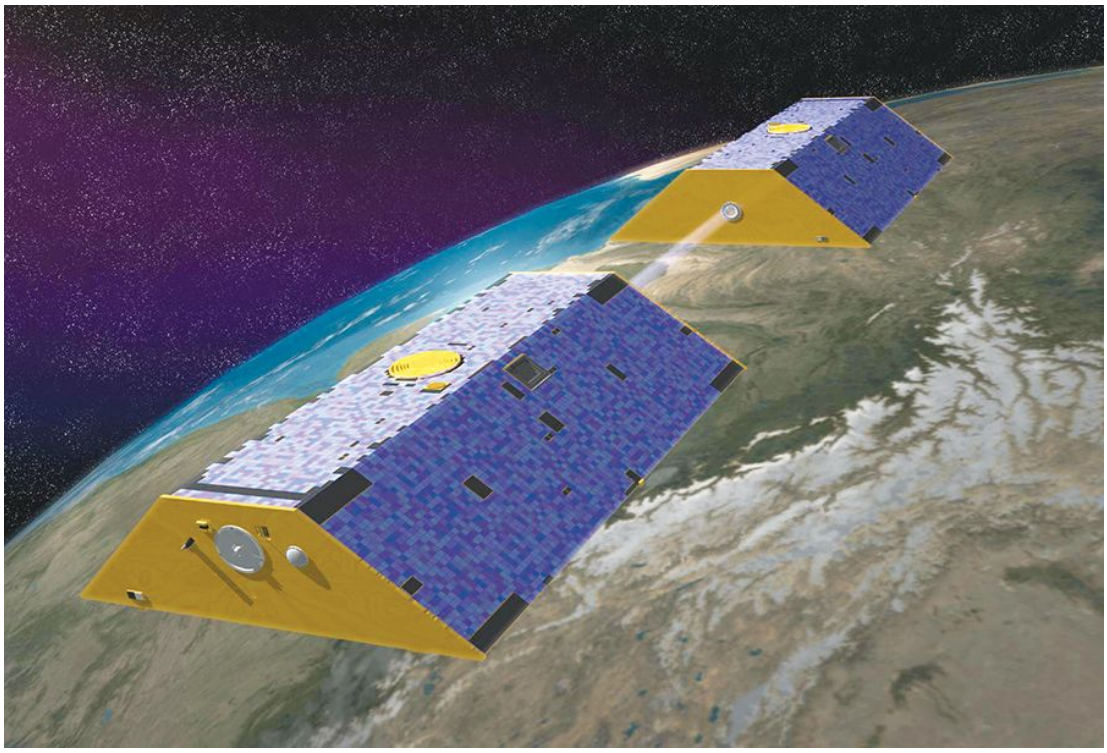


Fig. 24.8. GRACE: Two satellites, which track each other with a beam of microwaves, are pushed together by blue tendex lines and stretched apart by red tendex lines. The tendex lines, from the Earth below, are not shown.

In my interpretation of *Interstellar*, most of the gravitational anomalies that Professor Brand's team measures are sudden and unexpected changes in the patterns of tendex lines above the Earth's surface, changes that occur for no obvious reason. The rocks and oil in the Earth's crust are not moving. The melting of ice sheets is much too slow to produce these quick changes. People see no new gravitating masses coming near the gradiometers. Nevertheless, the gradiometers report changing tidal patterns. Falling dust accumulates in radial lines. Cooper sees the coin plunge to the floor.

The members of Professor Brand's team monitor these changing patterns and eagerly record Cooper's observations. Their trove of data becomes grist for the Professor's quest to understand gravity, a quest that centers on the Professor's equation.